

Tutorial 1: 2D Geometry

Homework: Exercises marked with red stars, with points adding to 10.

Curves in $\mathbb{R}^2$

Planar curves

There are two major ways to describe a curve in the plane: with an **equation**, and with a **parametrization**. An equation should contain two variables, usually called x and y . Every point (x, y) , for which x and y satisfy the equation then is a member of the curve. For example, the equation $x^2 + y^2 = 1$ describes the curve that contains all points for which the distance of (x, y) to the origin is 1, hence it describes the unit circle.

A parametrization is a function that accepts a variable, called a **parameter**, that maps this parameter to a point in the plane. In this way, every value of the parameter corresponds to a point in the plane. You build the curve by collecting these points for all possible values of the parameter. An example is the function $\mathbf{r}$ that maps a value of θ to the point $\mathbf{r}(\theta) = (\cos \theta, \sin \theta)$. Here θ is the parameter. Note that $\mathbf{r}(\theta)$ lies on the unit circle (why?).

If you allow θ to have values in the range $[0, 2\pi)$ (called the **domain** of $\mathbf{r}$), then $\mathbf{r}(\theta)$ assumes all points of the unit circle exactly once. You may choose a different domain, and still obtain the same figure. For example, the range $(-\pi, \pi]$ will also produce the full unit circle. Be careful though. If you make the domain too large, some points will be produced more than once. This might cause problems while plotting the curve. Lines will become thicker than intended. And if you make the domain too small, some points will not be produced at all. In the example, the range $[0, \pi]$ will produce a full circle, but just the part above the X-axis. On the other hand, that might just be what you wanted...

EXERCISES

1 *String art*

Usually, a line is defined as an equation of the form

$$\ell: y = ax + b, \quad (1)$$

where a is the **slope** of the line, and b is the **y -intercept** (the intersection of the line with the y -axis). An alternative description is the **intercept equation**:

$$\ell: \frac{x}{p} + \frac{y}{q} = 1. \quad (2)$$

The numbers p and q are the x and y -intercepts of ℓ .

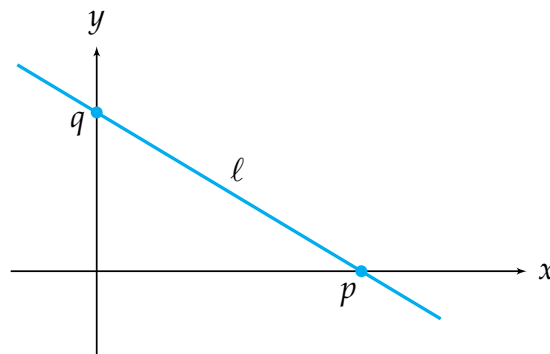


Figure 1: Intercepts of a line.

- (a) The intercept equation can only be defined for lines that are not parallel to the x or y -axis. Explain why.
- (b) Write equation (2) in the form described in (1). What is the slope of ℓ ?

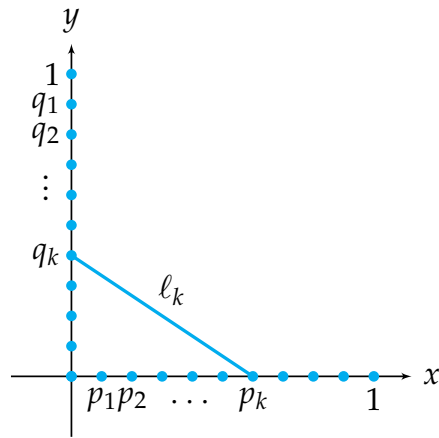


Figure 2: String ℓ_k connecting nails p_k and q_k .

Divide the intervals from 0 to 1 on both x and y -axis in 10 sub-intervals. See figure 2. The blue dots indicate where you put the nails.

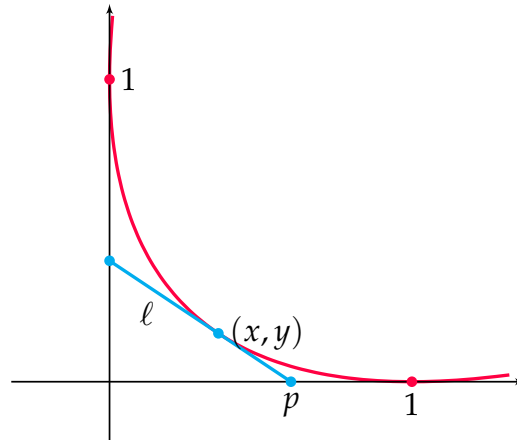
- (c) How many nails are there on the x -axis?
- (d) Let k be an integer in the range $1, 2, \dots, 9$. String ℓ_k connects the nail at p_k to the nail at q_k . Calculate p_k and q_k .
- (e) Calculate an equation for ℓ_k in the form $y = ax + b$.

2 The envelope

The envelope of the lines that constitute string art is a parabola. For the standard case (nails from 0 to 1 on both x and y -axis), this parabola is described by the equation

$$E: \sqrt{x} + \sqrt{y} = 1. \quad (1)$$

In general, envelope curves are difficult to calculate. Even if you know the envelope, it is hard to show that lines are tangent to envelope. In this exercise we will show a weaker result, namely that every string intersects E in one point. As a consequence, we can use this point to parametrize E .

Figure 3: String ℓ with x -intercept p .

In order to simplify equations, we use the notation p for the x -intercept of a string. We may safely assume that $0 < p < 1$.

- (a) As in the previous exercise, the shape we want to construct is such that the distances from the attachment points of the line (the 'nails') sum up to 1. Calculate the intercept equation for the blue line ℓ given as in the figure above.

The square roots are difficult to handle. The trick is to, for some specific point (x, y) in the envelope plotted in red in the figure, to replace x and y in (1) and in the equation for ℓ (since this point belongs to it) by new variables u and v . Define

$$\begin{cases} u &= \sqrt{x}, \\ v &= \sqrt{y}. \end{cases}$$

- (b) Express the equation (1) for E and the equation for ℓ in u and v .
- (c) Use the equation for E to write v as an expression in u . Then substitute this in the equation for ℓ . This gives a quadratic equation in u . Solve this equation. Then solve v .
- (d) Calculate x and y , using the result (c).

The result is a point with coordinates that depend on the constant p . By varying p over a range of values, we obtain a **parametrization** of the envelope E .

- (e) Use this parametrization to plot E . Also plot some string lines to produce your own string art.

3 The Spirograph



Figure 4: picture of an original Spirograph set (source: Wikipedia)

In 1965, the British engineer Denys Fisher started with the distribution of the Spirograph, a toy with which you can draw beautiful curly curves. The Spirograph was an instant hit, but after a few years, its popularity gradually waned. In 2013 the Spirograph was relaunched, and nowadays it is still available for purchase. There are several online versions available, for example [this one](#) by Nathan Friend.

The main parts of the Spirograph are plastic cog-wheels. One of the large wheels with a hole in the middle is attached with pins to a sheet of paper. Then a smaller wheel is placed either against the inside or the outside of the large fixed gear. With a ballpoint pen, you draw a line while rolling the small wheel around or inside the large wheel. Since the gears have a whole number of cogs, after a while, you will reach the starting point of the curve. If you roll the little wheel inside the large wheel, the curve that the Spirograph draws is called an **hypotrochoid**. If you roll the little wheel

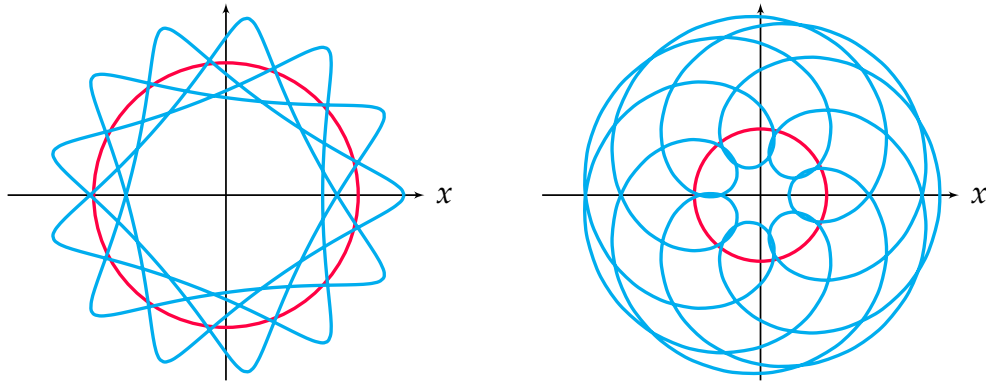


Figure 5: a hypotrochoid (left) with $R = 1$, $r = 9/13$ and $d = 1.5r$ and an epitrochoid (right) with $R = 1$, $r = 4/7$ and $d = 2r$.

along the outside of the large wheel, the curve is called an **epitrochoid**¹. In figure 5 you see examples of a hypotrochoid and an epitrochoid. The red circle is the unit circle. The parametric equations for the hypotrochoid are

$$\begin{cases} x = (R - r) \cos \theta + d \cos \left(\frac{R - r}{r} \theta \right), \\ y = (R - r) \sin \theta - d \sin \left(\frac{R - r}{r} \theta \right). \end{cases}$$

The parameter is θ . The number R is the radius of the fixed circle, and r is the radius of the rotating circle. The number d is the distance between the center of the rotating circle and the point that traces the hypotrochoid.

Write a computer program with which you can draw epitrochoids and hypotrochoids. Let θ run from 0 to $\theta_{\max}$. The main problem is to figure out when to stop: you want the whole curve, but you also do not want to plot part of the curve twice. Find a formula for $\theta_{\max}$, depending on R and r , such that the parametrization describes all points exactly once.



4 Ovals of Cassini

The **Cassini Oval** is defined as the set of points P such that the product of the distances $|PP_1|$ and $|PP_2|$ to given points P_1 and P_2 is constant.

¹“trochos” is Greek for wheel, “hypo” means under or below, “epi” means around or over.

Usually, the points P_1 and P_2 (called the **foci** of the oval) are distributed symmetrically on the X -axis. For simplicity, we assume that P_1 and P_2 have distance 1 to the origin, so $P_1 = (1, 0)$ and $P_2 = (-1, 0)$. Traditionally, the product $|PP_1| |PP_2|$ is set equal to b^2 , where $b > 0$.

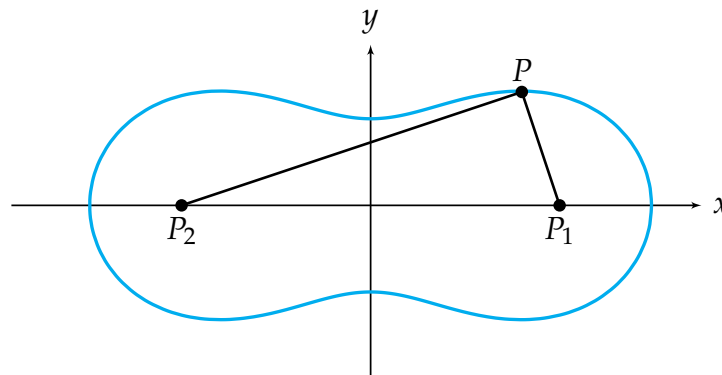


Figure 6: A Cassini oval with $b = 1.1$

Figures 6 show Cassini's oval with $b = 1.1$. Figures 7 demonstrates that the name 'oval' is a misnomer, since the curve now consists of *two* closed loops.

2

- (a) Let $P = (x, y)$. Express the equation $|PP_1|^2 |PP_2|^2 = b^4$ in x, y and b . Simplify as much as possible.

We want to parametrize the oval, using polar coordinates. For this we need to rewrite the equation in a specific form, namely

$$(x^2 + y^2)^2 + \dots (x^2 - y^2) + \dots = 0. \quad (1)$$

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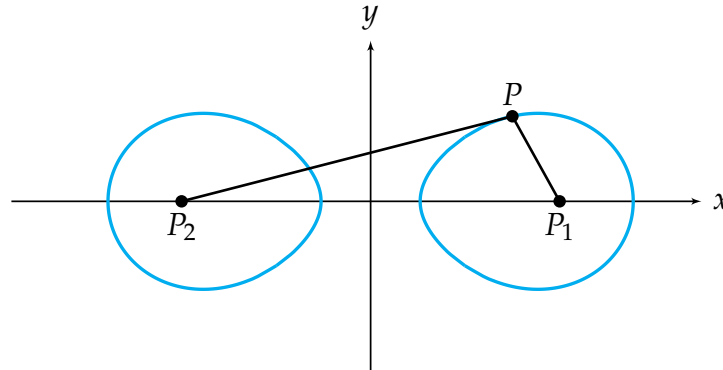
- (b) In equation (1), fill in the dots. The formulas that you use shouldn't contain x or y . Only constants and b are allowed.

1

- (c) Find the intersection points of Cassini's oval and the y axis. For which values of b are there two intersection points? For which values of b is there exactly one intersection point? For which values of b are there no intersection points?

1

- (d) Find the intersection points of Cassini's oval and the x axis. Show that there are always at least two intersection points. For which values of b are there two intersection points? For which values of b are there three

Figure 7: A Cassini oval with $b = 0.965$

intersection point? For which values of b are there four intersection points?

Now we are going to replace x and y by polar coordinates. Recall that for polar coordinates we have

$$\begin{cases} x = r \cos \theta, \\ y = r \sin \theta. \end{cases}$$

- (e) Rewrite the expressions $x^2 + y^2$ and $x^2 - y^2$ in polar coordinates. Simplify as much as possible. Then replace $x^2 + y^2$ and $x^2 - y^2$ in equation (1) by the expressions that you just derived.
- (f) In the equation that you obtain, replace r^2 by w .

Our goal is to write r as a function of θ . If all is well, you now have a quadratic equation in w , that you can solve using the quadratic formula.

- (g) Write w as function of θ .
- (h) Then write r as function of θ .

5 Plot the space Lissajous curve given by the parametric equations

$$\begin{cases} x = \sin\left(t - \frac{\pi}{2}\right) \\ y = \sin(9t) \\ z = \sin(10t) \end{cases}$$

Find a suitable domain for the parametrization: make sure that the curve is closed, but plot all points only once. Do you notice anything?

6 Supercircles

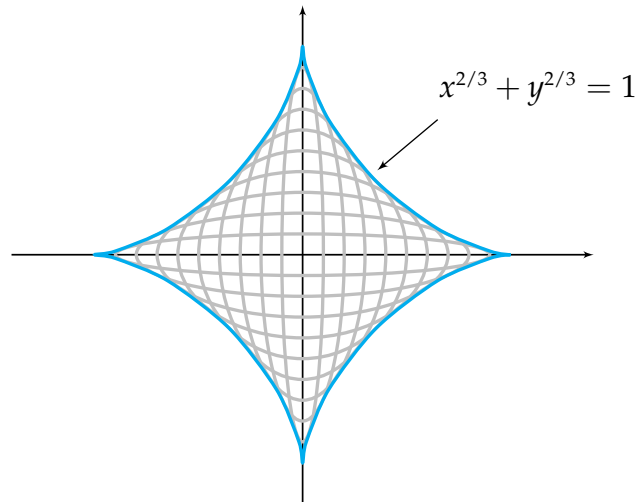


Figure 8: The astroid

In the lectures we saw that the envelope of a family of nested ellipses is the **astroid**, defined by the equation

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1.$$

The astroid is itself member of a family of curves called **supercircles**, which are defined by

$$S_\mu: |x|^\mu + |y|^\mu = 1,$$

where μ is a positive real number. The absolute value bars are necessary because for certain values of μ (e.g. $\mu = \frac{1}{2}$), the powers x^μ and y^μ are undefined.

(a) Does $S_{1/2}$ remind you of something?

(b) Sketch S_1 and S_2 in one plot. Don't use a computer.

In order to plot a supercircle, it would be handy to have a parametrization. We are going to try a variant of polar coordinates. Define

$$(*) \quad \begin{cases} x = \pm \cos^\alpha(\theta), \\ y = \pm \sin^\alpha(\theta), \end{cases}$$

for some yet to determine constant α . The “ $\pm$ ”-signs are necessary, because without them only the part in the first quadrant would be plotted. In fact, equations (*) define *four* parametrizations.

- (c) Find the correct value of α such that (*) is a parametrization of S_μ .
- (d) Using (*) and the result of (c), plot a family of supercircles in one figure.

The supercircles S_0 and S_∞ are undefined. However, by taking limits on can give sensible definitions for them.

- (e) How would you define S_0 ?
- (f) How would you define S_∞ ?

Contours

Contours in the plane

As mentioned in the previous section, planar curves can be described with an equation. There is a nice method to produce equations, namely by regarding them as contours of a function. For this, you need a function of two variables, say $f(x, y)$. Now choose a constant value C , and collect all points (x, y) for which $f(x, y) = C$. The set that you obtain is called the **contour of f at level C** . Very often, contours are curves. But this is not always the case. Sometimes a contour consists of two points, sometimes it's the empty set. In fact, anything can happen.

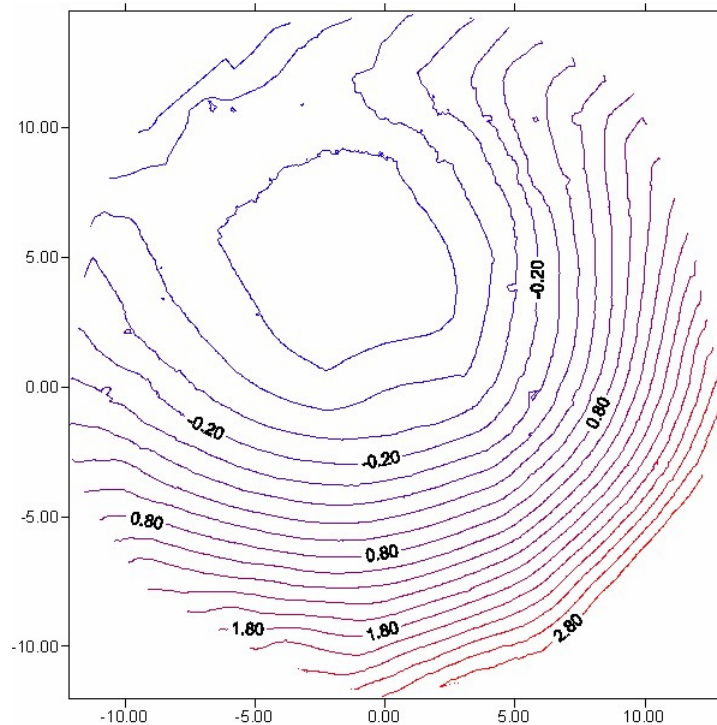


Figure 9: contour map of a small part of Mars, surveyed by the Mars Exploration Rover 'Opportunity'. Photo: NASA/JPL/Ohio State University.

If you choose a different value for C , you get a different contour. Contours have an important property: they cannot intersect (can you figure out why?). In figure 9 you can see some contours produced by the Mars rover 'Opportunity'. The level value is used as a label in some of the contours.

What a contour does is that it connects all points in the plane that have equal function value (namely C). That's why some contour names are prefixed by the word 'iso', which originates from the Greek word 'isos', meaning 'equal'. For example, isobars are curves that connect points with the same pressure. You see them a lot in weather charts.

EXERCISES

- 7 Define the function $f(x, y) = x^2 + y^2$. Are all contours circles? What is the radius of the circle $f(x, y) = C$?



8 Define the function of two variables

$$f(x, y) = \frac{2y^2}{1 - x^2}.$$



(a) Describe and draw the contours $f(x, y) = C$ for $C = 2$ and for $C = 1/2$.



(b) Describe the contours for all values of C (positive and negative).

9 Study the contour map made by Opportunity (fig. 9).

(a) Is what you see a hill or a crater?

(b) Estimate the lowest/highest level on the map.

10 The ovals of Cassini (see exercise 4) are contours of a function $f(x, y)$. Find this function, then make a contour plot of this function.

Regions, images, and transformations

EXERCISES

11 Consider the figure



and think of it as a function $I : \mathbb{R}^2 \rightarrow [0, 1]$ that assigns the number 1 to the inside of the object (blue) and 0 to all other points. Sketch the image produced by the function $I(f(x, y))$, when f is:

- A positive rotation of $\pi/2$ radians (90 degrees), that is

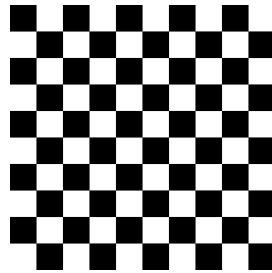
$$f(x, y) = (-y, x).$$

- The function given by

$$f(x, y) = (x/2, y).$$

Do you notice a pattern in the two results?

- 12 Write a formula for a function $I : \mathbb{R}^2 \rightarrow [0, 1]$ which produces the image of a checkerboard, like:



with white being the value 1, and black the value 0.

Complex numbers and complex functions

If we represent two points in Cartesian coordinates (x_1, y_1) and (x_2, y_2) as the complex numbers $x_1 + y_1i$ and $x_2 + y_2i$ where $i = (0, 1)$ is the imaginary unit, their complex product is written as:

$$(x_1 + y_1i)(x_2 + y_2i) = (x_1x_2 - y_1y_2) + (y_1x_2 + x_1y_2)i,$$

and we have

$$i^2 = -1, \quad \text{and} \quad (x + yi)^{-1} = \frac{x - yi}{x^2 + y^2}.$$

When the complex numbers $z_1, z_2 \in \mathbb{C}$ are in polar form $z_1 = r_1e^{i\theta_1}$ and $z_2 = r_2e^{i\theta_2}$, we have instead

$$z_1z_2 = (r_1e^{i\theta_1})(r_2e^{i\theta_2}) = (r_1r_2)e^{i(\theta_1+\theta_2)}, \quad \text{and} \quad (z_1)^{-1} = \frac{1}{r_1}e^{-i\theta_1}.$$

Complex multiplication and division allow us to easily write transformations of the plane to achieve interesting effects. For example, Möbius transformations are those of the form

$$f(z) = \frac{az + b}{cz + d} \quad \text{for some } a, b, c, d \in \mathbb{C},$$

and with them one can transform lines into circles and circles into lines.

You can try out a calculator like [this one](#) to plot the effect of complex functions on basic patterns or your own uploaded images.

EXERCISES

- 13 Say we want to rotate the plane by an angle $\pi/4$ radians (45 degrees) counterclockwise around the point $(1, 1)$. This can be accomplished with a complex linear transformation $f = az + b$. What are the complex numbers $a \in \mathbb{C}$ and $b \in \mathbb{C}$?
- 14 Consider the inversion $f(z) = 1/z$ and the horizontal line given by $y = 1$, which is the set of points

$$L = \{(x, 1) \in \mathbb{R}^2 \mid x \in \mathbb{R}\} = \{x + i \in \mathbb{C} \mid x \in \mathbb{R}\}.$$

What does L transform to when applying f to each of its points? Justify your answer.

Polygons

Polytopes are geometrical constellations of arbitrary dimension with flat sides. With ‘flat’ we mean lines and planes. In general, the sides of a polytope are polytopes themselves, but with lower dimension: a cube is a three-dimensional polytope with squares as sides, which themselves are two-dimensional polytopes. And squares have one-dimensional sides (line segments). A two-dimensional polytope is called a **polygon**. Examples are triangles, squares, and stars. The sides of a polygon are essentially line segments. The side of a polygon is sometimes referred to as an **edge**. These edges are conjoined in a point, called a **vertex**. An **n -gon** is a polygon with

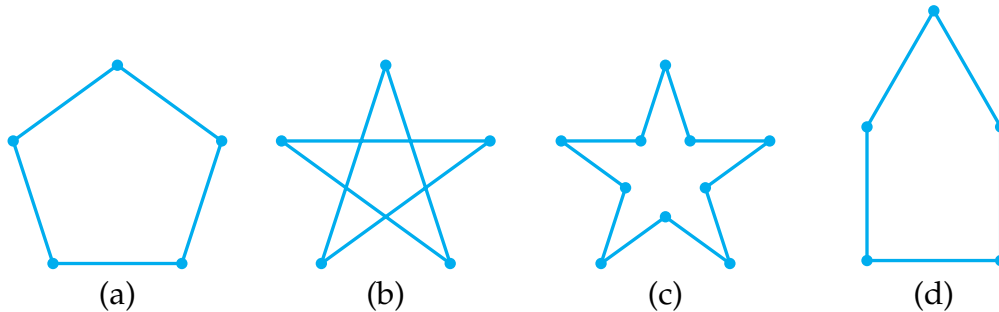


Figure 10: examples of some polygons

n vertices. If all edges have the same length, then the polygon is called **equilateral**. If all angles between two edges meeting in one vertex are the same, then the polygon is called **equiangular**.

EXERCISES

15 For each of the n -gons in figure 10, answer the following questions:

- What is n ?
- Is the polygon equilateral?
- Is the polygon equiangular?

For polygon (d) you may assume that the angle at the top is 60° .

16 A **regular n -gon** (notation: $\{n\}$) is a convex polygon that is equiangular and equilateral. A square is a regular 4-gon. In figure 10, polygon (a) is a regular 5-gon.

The construction of a regular n -gon can be done by rotating a starting vertex P_1 about a fixed point through angles $\frac{360^\circ}{n}$, resulting in n vertices $P_1, P_2, \dots, P_n$. The successive joins $P_1P_2, P_2P_3, \dots, P_{n-1}P_n$ are the sides of the n -gon.

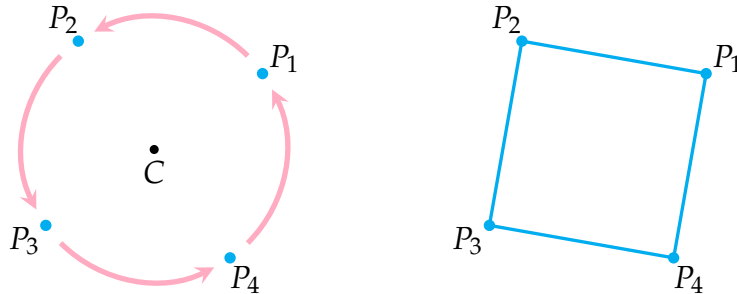
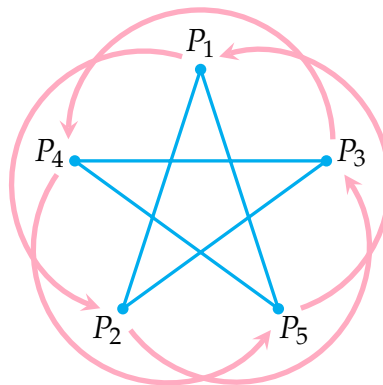


Figure 11: square obtained by rotating P_1 four times over an angle of 90° about C .

A so called **star polygon** is constructed by allowing n to be a rational number $\frac{p}{d}$, resulting in p vertices obtained by rotating over an angle of $\frac{360^\circ}{p/d}$.

For example, this is a $\left\{\frac{5}{2}\right\}$ star:



- Construct the following stars: $\left\{\frac{8}{3}\right\}$, $\left\{\frac{10}{3}\right\}$ and $\left\{\frac{12}{5}\right\}$.
- Find the angle at a vertex for $\{5\}$ and for $\left\{\frac{10}{3}\right\}$.
- A ray emanating from the center of rotation (not passing a vertex or an intersection point of two sides) will cross d of the p sides. Why is that? (We call d the **density** of $\left\{\frac{p}{d}\right\}$.)

Hints

- 1
 - (a) What happens with p if ℓ is parallel to the x -axis? And what happens with q if ℓ is parallel to the y -axis?
 - (c) The number of nails is 1 more than the number of sub-intervals.
 - (d) Note that the numbers p_k count from left to right, while the number q_k count downwards.
 - (e) Calculate the intercept equation of ℓ_k first.

- 2
 - (a) The y -intercept of ℓ is $1 - p$.
 - (c) It helps if you get rid of the denominators p and $1 - p$ in the quadratic equation. Multiply left- and right handside with $p(1 - p)$.

- 3 Hypo- and epitrochoids are closed curves whenever R/r is a rational number.

- 4
 - (c) In equation 1, set $x = 0$.
 - (d) In equation 1, set $y = 0$.
 - (e) Use $\cos^2 \theta + \sin^2 \theta = 1$ (Pythagoras' theorem) and the doubling formula for cosine: $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$.

- 6
 - (b)
 - For S_2 you don't need the absolute value bars.
 - For S_1 : start with the equation for S_1 without absolute value bars. Then use reflections to complete the figure.
 - (c) Use $\cos^2 \theta + \sin^2 \theta = 1$.
 - (d) To get a nice looking set of supercircles, use an exponential sequence for the μ -parameter, e.g. $\mu = (4/3)^n$, with $n = -2, -1, \dots, 6$.
 - (e) Plot S_μ with μ almost equal to 0.
 - (f) Plot S_μ with μ very large.

- 8 It might help to manipulate the resulting equation to get rid of the fraction.

- 9 (b) What is the distance in height between two successive contours?
- 10 You can choose the constant b to be the level value. Better is to use b^4 , then you avoid taking the 4-th root.
- 11 Remember: first transform, then assign the value according to the original image.
- 12 Two useful functions one can use are:
- The floor function for real numbers, which assigns to a number t the closest integer below t ,
- $$\lfloor t \rfloor = \max\{m \in \mathbb{Z} \mid m \leq t\}.$$
- The remainder of dividing an integer by two,
- $$k \pmod{2} = \begin{cases} 0 & \text{if } k \text{ even,} \\ 1 & \text{if } k \text{ odd.} \end{cases}$$
- 13 Remember that adding a complex number $b = x_b + y_b i$ represents translation, and multiplying by $e^{i\theta}$ is a rotation with angle θ .
- 14 Check the effect of f on the three points $z_1 = -1 + i$, $z_1 = i$ and $z_1 = 1 + i$, and remember that f is a Möbius transformation.
- 15 Intersections of edges do not count as vertices.